

What is **A2A** Protocol?

Introduction

What you've learned so far

In this lesson, all your agents run in the **same application**:

```
Python
# All agents in one place
coordinator = LlmAgent(
    sub_agents=[billing_agent, technical_agent] # Local agents
)
```

This works great for most use cases. But what about enterprise scenarios where agents need to communicate across services?

What is the A2A Protocol?

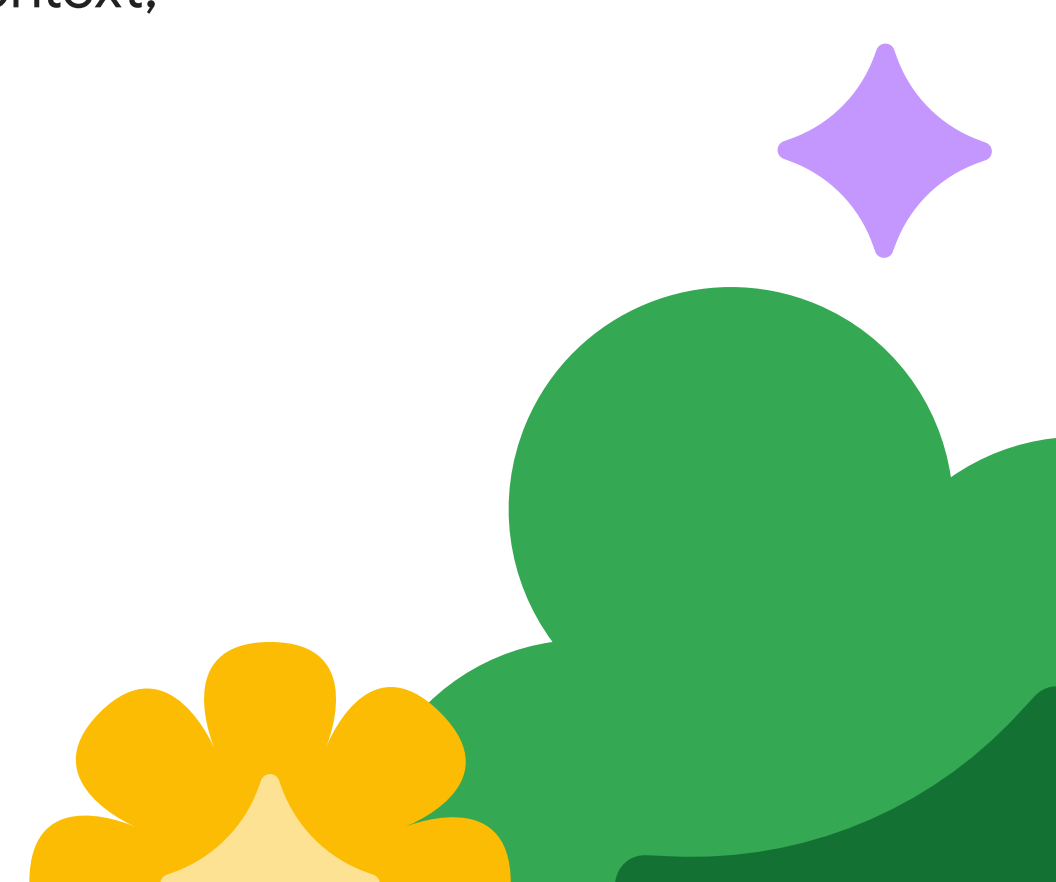
Agent-to-agent communication standard

The A2A (Agent2Agent) protocol is an open standard for agents to communicate across network boundaries.

From the official A2A website:

“A2A is a standard that allows agents to communicate with each other, share context, and work together across services and organizations.”

Key concepts:

- **Expose:** Make your agent available as a network service
 - **Consume:** Call remote agents like they're local
 - **Agent Card:** A contract describing what an agent can do
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When you might need A2A

Use cases for distributed agents

A2A is designed for enterprise scenarios:

Scenario	Why A2A?
Microservices	Agents as independent services that scale separately
Cross-team	Different teams maintain different agents
Cross-organization	Third-party agents you want to integrate
Different languages	Agents in Python, Java, etc. communicating

When you DON'T need A2A

For most learning and development, local agents are sufficient:

Your Situation	Recommendation
Learning ADK	Use local <code>sub_agents</code>
Single application	Use local <code>sub_agents</code>
Same team, same codebase	Use local <code>sub_agents</code>
Performance-critical	Use local <code>sub_agents</code> (no network overhead)

